

SCHISM-AED Compilation on Athena HPC

Verified installation procedure for DETSI / catchmodel project users

Prepared by Dongfang Liu, Aditya Singh and Badin Gibbes

Status	Compilation completed successfully
Installation date	2 July 2026
Project location	/proj/catchmodel/schism_aed
Executable	/proj/catchmodel/schism_aed/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL
Scheduler	PBS Pro
Runtime validation	Not yet completed; a small SCHISM-AED test case is still required

Important: This document records a successful compilation and link check. It does not yet confirm that a complete SCHISM-AED simulation runs successfully on Athena. A model test should be completed before the installation is treated as fully operational.

1. Overview

Athena differs from UQ Bunya in two important ways: Athena uses PBS Pro rather than Slurm, and the available MPI implementation is Intel oneAPI MPI rather than OpenMPI. The Bunya SCHISM-AED guide was therefore adapted for Athena while retaining the software versions supplied by Aditya.

2. Tested software environment

Component	Tested version	Athena source
GCC / GFortran	13.2.0	Spack
NetCDF-C	4.9.2	Spack
NetCDF-Fortran	4.6.0	Spack
CMake	3.30.5	Spack
MPI	Intel oneAPI MPI 2021.14	Athena dependency environment
Scheduler	PBS Pro	Athena

3. Request an interactive compute node

Do not compile on an Athena login node such as athena61. Use the interactive PBS queue.

```
qsub -I -q interq \
-l select=1:ncpus=8:mem=32gb \
-l walltime=02:00:00
```

After allocation, confirm that the hostname has changed and that a PBS job ID is available:

```
hostname
echo "$PBS_JOBID"
```

Expected result: The hostname should be a compute node, for example athena53, rather than athena61.

4. Initialise Spack and load the tested environment

```
source /opt/sw/spack/hpc/startHPC.sh
```

Load the exact installed package hashes used in the verified build:

```
spack load /ln6ek2b # gcc@13.2.0
spack load /jclrs2i # netcdf-c@4.9.2
spack load /gibamqu # netcdf-fortran@4.6.0
spack load /3dyxf6h # cmake@3.30.5
```

Verify the environment:

```
gcc --version | head -1
gfortran --version | head -1
nc-config --version
nf-config --version
cmake --version | head -1
which mpif90
mpif90 -show
spack find --loaded
```

Compatibility check: The verified build used GCC/GFortran 13.2.0 with Intel MPI 2021.14. The MPI wrapper was tested by compiling and running a two-process Fortran MPI program.

5. Prepare shared project storage

The source code and executable are stored in project space. Simulation cases and large outputs should be stored in scratch space.

```
umask 0002
mkdir -p /proj/catchmodel/schism_aed
mkdir -p /scratch/catchmodel/$USER
```

Permissions: Use umask 0002 in the shared project directory so members of the catchmodel group can read and modify project files.

6. Clone AED_Tools and fetch sources

```
cd /proj/catchmodel/schism_aed
git clone https://github.com/AquaticEcoDynamics/AED_Tools.git
cd AED_Tools
chmod -R g+rwX,o-rwx .
```

Fetch the public AED libraries and SCHISM source:

```
./fetch_sources.sh libaed schism
```

7. Ensure the required SCHISM commit is used

In the tested AED_Tools version, a fresh SCHISM clone did not immediately check out the commit required by the SCHISM-AED patch. Run the SCHISM fetch step a second time after the first clone:

```
./fetch_sources.sh schism
```

Verify the required and current commits:

```
cat schism-aed/gitlog-schism
git -C schism rev-parse HEAD
```

Verified SCHISM commit: e59606c8ef9cc049ef616b8c691e0cf363e083fc

Confirm that the AED source link exists:

```
ls -ld schism/src/AED
```

8. Adapt the build scripts for Athena

Back up the two build files before editing:

```
cp build_env.inc build_env.inc.original
cp build_schism.sh build_schism.sh.original
```

8.1 Disable GD for the SCHISM build

```
sed -i '"/build_schism.sh"/,/,;/s/NEED_GD=true/NEED_GD=false/' build_env.inc
```

The SCHISM block should contain `NEED_GD=false`.

8.2 Make AED_Tools recognise Intel MPI

The original script searches for `mpi_info` and assumes OpenMPI. Athena provides Intel MPI through `I_MPI_ROOT`. Replace lines 151-166 of `build_env.inc` with the following block:

```
sed -i '151,166c\
# Detect an available MPI installation: Intel MPI on Athena or OpenMPI elsewhere\
if [ "$NEED_OMPI" = "true" ] ; then\
  OMPIBASE=\
  if [ -n "$I_MPI_ROOT" ] && [ -x "$I_MPI_ROOT/bin/mpif90" ] && [ -x "$I_MPI_ROOT/bin/mpicc" ] ; then\
    OMPIBASE="$I_MPI_ROOT"\
    NEED_OMPI=false\
  else\
    BINDIR=`which mpi_info 2>/dev/null`\
    if [ "$BINDIR" != "" ] ; then\
      BINDIR=`dirname "$BINDIR"`\
      OMPIBASE=`dirname "$BINDIR"`\
      NEED_OMPI=false\
    fi\
  fi\
fi\
fi' build_env.inc
```

Check shell syntax before building:

```
bash -n build_env.inc && echo "build_env.inc syntax OK"
```

9. Compile SCHISM-AED

The build script defaults to `-j8`, matching the eight CPUs requested from PBS:

```
./build_schism.sh --with-aed 2>&1 | tee build_athena.log
```

Observed warning: Several legacy MPI calls produced argument type-mismatch warnings. These warnings did not prevent the core SCHISM-AED target from compiling.

10. Resolve the combine_output11 parallel build race

During the verified installation, the `-j8` build stopped because `combine_output11` and `combine_output11_MPI` attempted to generate the same `netcdf_var_names.mod` file concurrently. The key error was:

```
f951: Fatal Error: Cannot rename module file '././include/netcdf_var_names.mod0'
to '././include/netcdf_var_names.mod': No such file or directory
```

Resume from the existing build directory and rebuild the affected targets serially:

```
cd /proj/catchmodel/schism_aed/AED_Tools/schism/build

rm -f \
Utility/Combining_Scripts/CMakeFiles/combine_output11.dir/netcdf_var_names.f90.o \
Utility/Combining_Scripts/CMakeFiles/combine_output11.dir/netcdf_var_names.mod.stamp \
Utility/Combining_Scripts/CMakeFiles/combine_output11_MPL.dir/netcdf_var_names.f90.o \
Utility/Combining_Scripts/CMakeFiles/combine_output11_MPL.dir/netcdf_var_names.mod.stamp \
include/netcdf_var_names.mod \
include/NETCDF_VAR_NAMES.mod

make -j1 combine_output11
make -j1 combine_output11_MPL
make -j1 2>&1 | tee -a ../build_athena.log
```

Successful completion: The final serial build reached 100% and built the pschism target successfully.

11. Verify the executable and linked libraries

```
find schism/build/bin -maxdepth 1 -type f -perm -u+x -name 'pschism*' -ls
```

Verified executable:

```
/proj/catchmodel/schism_aed/AED_Tools/schism/build/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL
```

Check for missing shared libraries:

```
ldd schism/build/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL | grep "not found"
```

No output means no shared library is missing. Inspect key linked libraries:

```
ldd schism/build/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL \
| grep -Ei 'mpi|netcdf|hdf5|gfortran'
```

12. Install the executable and create a reusable environment script

```
mkdir -p /proj/catchmodel/schism_aed/bin
cp -p schism/build/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL \
/proj/catchmodel/schism_aed/bin/
```

Create /proj/catchmodel/schism_aed/load_athena_env.sh:

```
#!/bin/bash

umask 0002
source /opt/sw/spack/hpc/startHPC.sh

spack load /ln6ek2b
spack load /jclrs2i
spack load /gibamqu
spack load /3dyxf6h

export SCHISM_AED_ROOT=/proj/catchmodel/schism_aed
export SCHISM_AED_EXE=/proj/catchmodel/schism_aed/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL
chmod 775 /proj/catchmodel/schism_aed/load_athena_env.sh
```

For future jobs, load the environment with:

```
source /proj/catchmodel/schism_aed/load_athena_env.sh
```

13. Record the build provenance

```
mkdir -p /proj/catchmodel/schism_aed/docs

git -C /proj/catchmodel/schism_aed/AED_Tools rev-parse HEAD \
> /proj/catchmodel/schism_aed/docs/AED_Tools_commit.txt

git -C /proj/catchmodel/schism_aed/AED_Tools/schism rev-parse HEAD \
> /proj/catchmodel/schism_aed/docs/SCHISM_commit.txt

cp /proj/catchmodel/schism_aed/AED_Tools/build_athena.log \
/proj/catchmodel/schism_aed/docs/
```

Repository	Verified commit
AED_Tools	cfd97d55a7456df0223d0eb388c8edc7cbbffd7
SCHISM	e59606c8ef9cc049ef616b8c691e0cf363e083fc

14. PBS job submission note

Athena uses PBS Pro, while Bunya uses Slurm. Existing Bunya submission scripts cannot be used unchanged. A PBS job script must use #PBS directives and launch the executable using the MPI method supported by Athena. This manual intentionally does not provide a production run script because the runtime test has not yet been completed.

Next action: Transfer a known SCHISM-AED test case to Athena, confirm a short multi-process run, and then prepare a separate Athena PBS submission guide.

15. Troubleshooting summary

Symptom	Cause	Action
salloc, srun or sbatch not found	Athena uses PBS, not Slurm	Use qsub, qstat and qdel
ompi_info not found	Athena provides Intel MPI	Use the Intel MPI detection patch in Section 8.2
AED_Tools tries to build OpenMPI	build_env.inc cannot detect Intel MPI	Confirm I_MPI_ROOT and apply Section 8.2
SCHISM commit does not match gitlog-schism	Fresh clone does not perform checkout	Run ./fetch_sources.sh schism a second time
Cannot rename netcdf_var_names.mod0	Parallel race between two utility targets	Clean only the listed module products and rebuild serially
ldd reports not found	Required environment is not loaded	Source load_athena_env.sh before running

16. Final installation status

The SCHISM-AED source was compiled successfully on Athena, the pschism executable was produced, and all required shared libraries were resolved. The reusable environment script also passed a clean-shell test. A complete model runtime test remains outstanding.

Installed executable:

```
/proj/catchmodel/schism_aed/bin/pschism_AED_BLD_STANDALONE_SH_MEM_COMM_TVD-VL
```